RW–10575

Sub. Code
501504

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

***Elective* – II — MARINE ECOSYSTEM AND PRINCIPLES OF OCEANOGRAPHY**

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Comment on benthic biotic environment.

2. Define deep sea environment and its biota.

3. Differentiate between endangered species and edge species.

4. Define Southern Oscillation.

5. What is the Reynolds number of nekton?

6. Explain Somali current.

7. What are minor elements of sea water?

8. Define BOD and its significance.

9. How do you calculate gross primary productivity?

10. Relate climatic changes with Ozone depletion.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain any two factors and its impact on marine environment.

Or

(b) Write a note on Estuaries with example.

12. (a) Comment on ecosystem restoration and rehabilitation.

Or

(b) Explain the any two types of pollution in marine ecosystem.

13. (a) What is the significance of wave height and period?

Or

(b) Write short notes on tide generating forces.

14. (a) Give an brief account on sampling.

Or

(b) What are the effects of gases dissolved in seawater?

15. (a) List the factors involved in primary production.

Or

(b) Define Ozone depletion and its causes.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Short notes on Mangroves, coral reefs and coral bleaching.

17. Give the current status of marine biodiversity and its overexploitation.

18. Explain ENSO.

19. Give a detailed account on oceanographic instrument and its application in tackling natural calamities.

20. Explain marine food chain.

RW–10403

Sub. Code
501102

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

MICROBIOLOGY

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Contributions of Louis Pasteur.

2. Serotypes.

3. Pleomorphism.

4. Twitching motility.

5. Microbial antagonism.

6. Endemic disease.

7. RAPD.

8. ARDRA.

9. Define Mycorrhiza.

10. Pyridoxine.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Classify the microbes based on flagella.

Or

(b) Explain the structure of cell wall of gram positive bacteria.

12. (a) Differentiate flagella, axial filaments, fimbriae and pili.

Or

(b) Enumerate the ATP yield during prokaryotic aerobic respiration on one glucose molecule.

13. (a) Compare commensalism, mutualism and parasitism with suitable example for each.

Or

(b) Write short notes on Aspergillosis.

14. (a) Explain the principle of DGGE.

Or

(b) Write about the applications of T-RFLP.

15. (a) Briefly explain the production of antibiotics.

Or

(b) Define biodegradation. Name four organisms involved in biodegradation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain briefly the classification of bacteria with reference to Bergey's manual.

17. Compare and contrast the light-dependent and light independent reactions of photosynthesis.

18. Describe briefly different modes of disease transmission.

19. What is metagenomics and explain environmental gene surveys?

20. Briefly explain the industrial production of vitamins.

RW–10404

Sub. Code
501103

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

MOLECULAR BIOLOGY AND GENETICS

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. DNA gyrase.

2. Polyadenylation.

3. Attenuation.

4. Aporepressor.

5. Allele.

6. Centimorgan.

7. Prophage.

8. Horizontal gene transfer.

9. Monohybrid cross.

10. Trisomy.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Write short notes on nature of genetic code.

Or

(b) Compare the replication in prokaryotes and eukaryotes.

12. (a) Describe the regulatory elements of *trp* operon.

Or

(b) Write short notes on :

(i) Photoreactivation

(ii) Spontaneous mutation.

13. (a) Explain genetic linkage and crossing over.

Or

(b) Give an account on heat shock response.

14. (a) Write short notes on transposable elements.

Or

(b) Differentiate $F^+ \times F^-$ and $Hfr \times F^-$ mating.

15. (a) Give an account on Eugenics.

Or

(b) Explain Y-linked inheritance with a specific example.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Give a brief account of the different steps involved in transcription of prokaryotes.

17. Describe the regulation of gene expression with special reference to *lac* operon.

18. Explain the mechanism of SOS repair and its regulations.

19. Describe in detail about the mechanism of transposons.

20. Give an account on structural and numerical alterations of chromosomes.

RW–10405

Sub. Code
501104

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

CELL BIOLOGY

(2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Note on peroxisomes.

2. Nucleosomes.

3. Passive transport.

4. Role of Cdks in cell cycle.

5. What is meristem?

6. Gibberellins.

7. Role of caspase 3.

8. Define Hsp90.

9. Define oncogene.

10. Metastasis.

Part B $(5 \times 5 = 25)$

Answer the following.

11. (a) Write about the origin and function of Endoplasmic reticulum.

Or

(b) Write about chromatin organization.

12. (a) Write about intracellular protein sorting in mitochondria.

Or

(b) Write about regulations in cell cycle.

13. (a) Discuss the architecture of plant cell wall.

Or

(b) What are the different exo and endo stimuli drive differentiation in plant cell.

14. (a) Discuss proteasome function.

Or

(b) Explain cellular functions of chaperons.

15. (a) Explain any two aging theories.

Or

(b) Write about general characteristics of cancerous cells.

Part C (3 × 10 = 30)

Answer any **three** of the following.

16. Detail functions and organization of cytoskeletons.

17. Explain in detail about chloroplast structure and function.

18. Explain specific roles of hormones in cell differentiation in plants.

19. Narrate the classification chaperons and their role in regulation of cellular function.

20. Discuss theories regarding induction of cancer.

RW–10406

Sub. Code
501105

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

Lab I — ANALYTICAL BIOCHEMISTRY

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Spectrophotometry
2. Lowry's method
3. Km
4. ALP
5. pKa
6. pI
7. Extra cellular enzymes
8. Dialysis
9. Poisson distribution
10. Mode

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Write a short note on evaluation of beers law.

Or

(b) How will you determine a protein by using bradford's method?

12. (a) Write notes on subcellular fractionation.

Or

(b) Explain about the estimation of' serum cholesterol.

13. (a) How will you determine the pI values of amino acids?

Or

(b) How will you separate plant pigments by using TLC?

14. (a) Explain about the ammonium sulphate precipitation.

Or

(b) Write notes on ultrafiltration.

15. (a) How will you identify proteins by using 2D gels?

Or

(b) Write notes on Metrovariate statistics.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. How will you determine a carbohydrate by using DNSA method?

17. Explain the clinical significance and assay of enzymes SGOT, SGPT, ACP.

18. Derive Henderson-Hasselbach equation and explain how will you evaluate pKa values in acid-base titrations?

19. Discuss in detail about the principle and applications of HPLC.

20. Explain in detail about radioactive labeling and measurement of radioactivity.

RW–10407

Sub. Code
501106

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

LAB – II — MICROBIOLOGY

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Epifluorescence microscopy
2. Oil immersion
3. Dessication
4. Negative staining
5. Mixotrophs
6. Penicillin
7. Photobioreactor
8. MPN
9. Mechanism of urease test
10. DNA probes.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Principle of phase contrast microscopy.

Or

(b) Illustrate on enumeration of bacteria from water.

12. (a) Explain the techniques employed to measure microorganisms.

Or

(b) Explain the procedure used to purify the bacterial culture

13. (a) Illustrate the mechanism of antibiotic resistant.

Or

(b) Explain the screening and identification procedure for protease producing microbes

14. (a) Write a brief note on quorum sensing.

Or

(b) Give an account on different types of bioreactor.

15. (a) What are the biochemical tests used For the identification of *Vibrio cholera*?

Or

(b) Principle of IMVIC and oxidase tests.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. What are the safety measures followed in microbiology laboratory?

17. Describe the principle and methods involved to find out the motility of microorganisms.

18. Explain the methodology used to demonstrate the antibiotic resistance.

19. Explain water quality analysis using MPN method.

20. Give an account on molecular methods employed for bacterial identification.

———————

RW–10408

Sub. Code
501501

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

Elective I — BIOPHYSICS AND INSTRUMENTATION

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Van der waals forces
2. Hydrophobic interations
3. Primary structure of protein
4. Ramachandran plot
5. Radioactive isotope
6. Lyophilization
7. Sedimentation
8. Ethidium bromide
9. DAPI (4', 6-diamidino-2-phenylindole)
10. NMR.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Describe about ionic bond and peptide bond with suitable illustration.

Or

(b) Write a short note on pH.

12. (a) Explain about secondary and tertiary structure of protein.

Or

(b) Give a detailed account on Ramachandran plot.

13. (a) Explain the applications of radioactive isotopes in biological studies.

Or

(b) Describe about autoradiography.

14. (a) Write short notes on types of centrifuges.

Or

(b) Explain the principle and applications of ion exchange chromatography.

15. (a) Write a short note on X-ray diffraction method.

Or

(b) Give an account on light microscopy.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about the principles related to Thermodynamics.

17. Discuss in detail about structural polymorphisms of DMA.

18. How will you measure the radioactivity in a biological sample using gas ionization technique?

19. Describe the principle and application of agarose gel electrophoresis.

20. Explain in detail about scanning electron microscope.

RW–10409

Sub. Code
501301

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

BIOINFORMATICS

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. NCBI
2. Swiss PDB Viewer
3. Algorithm
4. Phylogenetic analysis
5. Chou Fasman method
6. Bulge loop
7. Contig assembly
8. EMBL
9. Docking
10. Homology modeling

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain about a biological database.

Or

(b) Write short notes on EMBOSS.

12. (a) Give a detailed account on database searching.

Or

(b) Describe about alignment viewers.

13. (a) How will you predict the secondary structure of protein?

Or

(b) Explain about tertiary structure prediction of protein.

14. (a) Write a short notes on designing of primer.

Or

(b) How do you predict a gene in a eukaryotic genomic DNA sequence?

15. (a) Write a short note on virtual screening.

Or

(b) Explain about molecular modeling.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss in detail about visualization of macromolecules using Rasmol

17. Give a detailed account on Global Vs Local alignment.

18. Explain in detail about RNA structure analysis and list the application of RNA structure.

19. Describe in detail about computing in proteomics.

20. Explain in detail about drug designing concepts.

———————

RW–10410

Sub. Code
501302

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

ANIMAL BIOTECHNOLOGY

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Dolly.
2. Vitrification.
3. Bacmid.
4. SV_{40} virus.
5. Pheromones.
6. Angiotensin.
7. Germ cells.
8. Magnetofection.
9. iPSCs.
10. Malignant tumor.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Describe artificial insemination. Mention its importance.

Or

(b) Describe the chemical methods of transferring genes in animal cell lines.

12. (a) Give an account on baculovirus in biocontrol.

Or

(b) Explain the retrovirus expression vector system with an illustration.

13. (a) Give an account on production of vaccines from animal biotechnology.

Or

(b) Elaborate on hormone receptors.

14. (a) Describe an application of phage display technology.

Or

(b) Differentiate *Ex vivo* and *in vivo* gene therapy.

15. (a) Write short notes on bioethics.

Or

(b) Describe Hematopoictic stem cell transplantation.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Write briefly various biological methods of transferring genes.

17. Describe the construction of recombinant animal viral vector for gene transfer into cell lines.

18. Explain signaling through G-protein coupled receptors.

19. Elucidate DNA based diagnosis of genetic diseases.

20. Give an account on application of stem cells therapy for the Parkinson's disease.

RW–10411

Sub. Code
501303

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

MARINE BIOTECHNOLOGY

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Tunicates.

2. Heavy metal pollutants.

3. Barophilic microorganism.

4. Extremozymes.

5. Aquaculture.

6. Gene probe.

7. Transposons.

8. Vector.

9. Auto inducers.

10. Pathogen.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Write short notes on marine organisms as sources of untapped resources.

Or

(b) Give a detail account on seaweeds as a source of polysaccharides.

12. (a) Give a detail account on hyperthermophilic microorganisms and their applications.

Or

(b) Give an account on unculturable bacteria and Dr. Craig Venter.

13. (a) Explain in detail about the characteristics and applications of GFP.

Or

(b) Write short notes on the vaccines for aquaculture.

14. (a) Write short notes on transgenic fishes with growth hormone.

Or

(b) Explain in detail about transposons in fishes.

15. (a) Give a detail account on quorum sensing.

Or

(b) Write short notes on bacterial cell-cell communication system.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about bioactive compounds from marine organisms

17. What are the biotechnological applications of extremozymes from extremophilic organisms?

18. Give a detail account probiotics bacteria and their importance in aquaculture

19. Discuss about transgenic fish technology

20. Explain in detail about quorum sensing inhibitor compounds and its role in expression of virulence genes among bacterial pathogens.

———————

RW–10412

Sub. Code
501304

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

LAB V — RECOMBINANT DNA TECHNOLOGY

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Lysozyme.
2. Proteinase K.
3. SSC buffer.
4. Autoradiography.
5. GC clamp.
6. Colony PCR.
7. GFP.
8. TaqMan probes.
9. TEMED.
10. PVDF.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Write the principle of genomic DNA isolation by CTAB method.

Or

(b) Describe the principle of plasmid DNA isolation by boiling method.

12. (a) Give an account on colony hybridization.

Or

(b) Briefly explain the method of sub-cloning in M13.

13. (a) Explain the principle of ARDRA.

Or

(b) Discuss site-directed mutagenesis.

14. (a) Elaborate on growth and maintenance of C. *elegans*.

Or

(b) Explain steps involved in Reverse Transcriptase PCR.

15. (a) Describe different types of Zymogram.

Or

(b) Elaborate the different methods for the purification of bioactive compounds.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write briefly the principle and method of total RNA isolation from gram negative bacteria.

17. Give a detailed account on construction of genomic DNA library with diagrammatic representation.

18. Explain different types of DNA finger printing methods and its applications.

19. Write a detailed note on methodology and applications of recombinant protein expression.

20. Give an account on down stream processing of bioactive compounds.

RW–10413

Sub. Code
501305

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

PLANT BIOTECHNOLOGY — LAB VI

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Callus.

2. Sterilization.

3. Organogenesis

4. ISSR.

5. Micropropagation.

6. Cryopreservation.

7. Genome.

8. Ti-plasmid.

9. Explant.

10. GUS gene.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) How will you prepare nutrient media for callus culture?

Or

(b) Describe sterilization of nutrient media.

12. (a) How will you establish and maintain a suspension culture?

Or

(b) Describe about shoot tip culture.

13. (a) Write short notes on synthetic seed preparation from intact regenerable explants of medicinal plants through gel entrapment.

Or

(b) Explain about plant conversion from synthetic seeds.

14. (a) Describe the principle and method of genomic DNA extraction from leaf.

Or

(b) Write notes on *Agrobacterium tumifaciens*.

15. (a) Describe PCR analysis of transformed plants.

Or

(b) Explain the principle and method of Biolistic transformation of food crops.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about processing of various explants for culture initiation.

17. Explain about RAPD and ISSR analysis of medicinal plants.

18. Discuss about low temperature storage and cryopreservation of plant genetic resources from endangered medicinal plants.

19. Illustrate cloning of abiotic responsive genes into binary vector.

20. Narrate *Agrobacterium rhizogens* mediated transformation of medicinal plants.

RW–10414

Sub. Code
501510

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

Elective III — IPR, BIOSAFETY AND BIOETHICS

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. WTO.

2. WIPO.

3. TRIPS.

4. Patent.

5. FlavrSavr.

6. Biopiracy.

7. Biological weapons.

8. Containment.

9. Bioethics.

10. ELSI.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Write notes on GATT.

Or

(b) Explain the roles of RCGM and GEAC.

12. (a) Write notes on the requirement of patentability.

Or

(b) Describe in detail about the traditional knowledge.

13. (a) Explain about the rules governing patents.

Or

(b) Describe about apatent related case (impeachment).

14. (a) Write short notes on guidelines for rDNA research activities.

Or

(b) Describe in detail about Cartagena Biosafety protocol.

15. (a) Write notes on animal rights.

Or

(b) Give an account on ethical issues related to research in stem cell cloning.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the establishment, functions and guidelines of WTO.

17. Describe about (a) Trademark (b) Trade secret (c) Copy right (d) Geographical indications

18. Narrate (a) patent application (b) Indian Patent Act, 1970 and recent amendments

19. Illustrate Good Laboratory Practices and Good Manufacturing Practices.

20. Discuss in detail about Ethical, Legal and Social Implications of human genome project.

———————

RW–10415

Sub. Code
501512

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

Elective IV — FUNCTIONAL GENOMICS

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Transcript.

2. Real Time PCR.

3. Isoelectric focusing.

4. NMR.

5. Mass spectrometry.

6. Phage display.

7. Pathogenicity.

8. C. *elegans*.

9. Phase I and phase II levels of drug discovery.

10. Pharmacogenetics.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Write short notes on differential gene expression.

Or

(b) Explain the principle and applications of DNA microarray.

12. (a) How will you analyze a protein by 2-D NMR?

Or

(b) Explain about Protein in microarray.

13. (a) Explain Yeast two hybrid system with clear illustration.

Or

(b) How will you use GFP to visualize proteins in live culture?

14. (a) Write short notes on functional genomic analysis of bacterial pathogens.

Or

(b) Give an account on the environmentally significant microorganisms.

15. (a) Write notes on high throughput screening for drug discovery.

Or

(b) How will you identify the drug targets?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the principle and applications of RT-PCR.

17. Discuss the analysis of protein by Mass spectroscopy.

18. Explain the followings with illustrations

    (a) Proteomics

    (b) two protein-protein interaction techniques.

19. How will you study human microbial pathogens using mice as a model animal?

20. Discuss about developing drugs using parmacogenetics approach.

RW–10571

Sub. Code
501201

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

IMMUNOBIOLOGY

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Adjuvants.

2. Haptens.

3. Cytokines.

4. Immunity.

5. Toll like receptor.

6. HLA tissue typing.

7. AIDS.

8. Organ transplantation.

9. Oncogene.

10. Hypersensitivity reactions.

Part B $(5 \times 5 = 25)$

Answer **all** the questions by choosing either (a) or (b).

Draw diagrams wherever required.

11. (a) Write short notes on the cells of the immune system.

Or

(b) Write short notes on innate immunity.

12. (a) Give a short note on T dependent antigens.

Or

(b) List the generation of antibody diversity.

13. (a) Write short notes on the classical and alternate pathways of complement systems.

Or

(b) Give an account on auto immune disorder.

14. (a) Explain the Immune response to viral in infectious diseases.

Or

(b) Explain the immune response to helminthes in infectious diseases.

15. (a) Write short notes on congenital immuno deficiencies.

Or

(b) Write short notes on edible vaccines.

Part C (3 × 10 = 30)

Answer any **three** questions.

Draw diagrams or charts wherever required.

16. Detail acquired immunity with respect to humoral and cell mediated immune response.

17. Explain the structure, distribution and function of immunoglobulins.

18. Write elaborate on active and passive immunization.

19. Explain the structure and its interaction with peptide of major histocompatibililv complex.

20. Illustrate type II and type IV hypersensitivity reactions.

RW–10572

Sub. Code
501202

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

RECOMBINANT DNA TECHNOLOGY

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Recombinant DNA.

2. PRIBNOW BOX.

3. Kozak sequence.

4. Cosmids.

5. Sticky ends.

6. Probe.

7. Dideoxy nucleotide.

8. Chromosome walking.

9. Cystic fibrosis.

10. Type I diabetes.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Give an account on alkaline phosphatase.

Or

(b) Give an account on thermostable DNA polymerases used in PCR.

12. (a) Discuss about Cosmid vectors.

Or

(b) List the desirable characteristics of a cloning vector.

13. (a) What are the uses of adapters and linkers?

Or

(b) Give an account on dot blot.

14. (a) Explain the importance of Dpn 1.

Or

(b) Give an account on chromosome jumping.

15. (a) How will you produce insulin for therapeutic use by using recombinant DNA technology?

Or

(b) Write notes on gene therapy.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write a detailed account on :
    (a) DNA dependent RNA polymerases
    (b) EXO III.

17. Explain in detail about YACs.

18. Give an account on immunological screening for expressed genes.

19. Explain in detail about high throughput Pyrosequencing.

20. Discuss in detail about
    (a) TPA
    (b) Hepatitis B vaccine.

RW–10573

Sub. Code
501203

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

PLANT MOLECULAR BIOLOGY

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Comment on promoter.

2. What are plastids?

3. What is vitrification?

4. What are the biotech techniques for plants biodiversity?

5. What is agroinfection and application?

6. Comment on super binary vector.

7. Mention two salinity stress responding genes.

8. Comment on ultra sonication gene transfer mechanism.

9. What is RNAi?

10. Explain the strategy used in the design of expression vectors.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Write short notes on organization of plant genome.

Or

(b) Briefly explain the promiscuous DNA.

12. (a) Explain the production and application of artificial seed.

Or

(b) Comment on *in vitro* conservasion of plants.

13. (a) Write short notes on co-integrate vectors.

Or

(b) Hairy root culture for mass production of secondary metabolites – Justify.

14. (a) Write a notes on transposon tagging.

Or

(b) Briefly explain the osmotic stress response in plants.

15. (a) Write a note on selectable markers in plant genetic engineering.

Or

(b) Explain the mechanism of herbicides resistance.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Discuss the targeting of nuclear encoded protein to chioroplast.

17. Elucidate the molecular markers

    (a) RAPD

    (b) AFLP

18. Elaborate on current approaches toward production of secondary plant metabolites.

19. Discuss PHB biosynthesis and its production using bioreactor.

20. Explain how the genetic engineering advances are useful for producing plants abiotic stress tolerance.

RW–10574

Sub. Code
501205

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Department of Biotechnology

Lab IV — IMMUNOTECHNOLOGY

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Hybridoma

2. Aminopterin

3. Apoptosis

4. Immunology

5. Vaccines

6. Immunoglobulin

7. Sub culture

8. RT-PCR

9. Immuno fluorescence

10. HIV.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

Draw diagrams wherever required.

11. (a) Explain the applications of monoclonal antibody in clinical diagnosis and treatment.

Or

(b) Explain the protocol of immunization.

12. (a) Write short note on immunodiffusion.

Or

(b) Give a note on haemagglutination.

13. (a) How will you determine the WBC count in blood?

Or

(b) Write short note on the differential count of WBC.

14. (a) How will you detect and prevent the contamination of cells?

Or

(b) Write short note on transient and stable transfection methods.

15. (a) Write short note on the diagnosit kits for identifying tuberculosis.

Or

(b) Explain RIA.

Part C (3 × 10 = 30)

Answer any **three** questions.

Draw diagram or charts wherever required.

16. Explain about monoclonal antibody production.

17. Demonstrate in detail about immunoelectrophoresis.

18. Describe the lymphocyte subset identification and enumeration by FACS.

19. How will you monitor the expression of immunoglobulin genes by RT-PCR in host-pathogen interaction using *C. elegans* as model organism?

20. Describe in detail about Western blotting.

———————

RW–10402

Sub. Code
501101

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Biotechnology

BIOCHEMISTRY

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Differentiate atomic number and mass number.

2. pH.

3. Oxidative phosphorylation

4. Glycolysis.

5. Integral membrane protein.

6. Ion channels.

7. Gluconeogenesis.

8. Coenzyme.

9. V max.

10. Enzyme immobilization.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Write short notes on

    (i) Van der waals forces

    (ii) hydrogen bonding

    (iii) hydrophobic interaction.

Or

(b) Explain about the different structures of glucose.

12. (a) Describe the structure and functions of ATP.

Or

(b) Write a short note on TCA cycle.

13. (a) Describe about fluid mosaic model of membrane.

Or

(b) Write short notes on active transport system.

14. (a) Give an account on pentose phosphate pathway.

Or

(b) Write short notes on metabolomics.

15. (a) Describe enzyme nomenclature and classification.

Or

(b) Write short notes on enzyme engineering.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Describe the composition, structure and functions of lipids.

17. Discuss in detail about electron transport chain.

18. Illustrate in detail about membrane proteins.

19. Explain in detail about fatty acid oxidation and biosynthesis of lipids.

20. Derive the Michaelis Menton equation with neat diagram.